

Set 9- Online_Learning_Activity.csv

9C. Date Conversion, Monthly Average & Line Chart

Program:

Convert Date

```
data$Login_Date <- as.Date(data$Login_Date)
```

Extract Month

```
data$Month <- format(data$Login_Date,"%Y-%m")
```

Average Quiz Score Per Month

```
monthly <- data %>%
```

```
group_by(Month) %>%
```

```
summarise(Avg_Quiz=mean(Quiz_Score))
```

Moving Average

```
monthly$MovingAvg <- rollmean(
```

```
monthly$Avg_Quiz,
```

```
k=2,fill=NA,
```

```
align="right")
```

Line Chart

```
ggplot(monthly,aes(x=Month,y=Avg_Quiz,group=1))+
```

```
geom_line(color="blue",size=1)+
```

```
geom_point(size=3,color="red")+
```

```
geom_line(aes(y=MovingAvg),
```

```
color="green",
```

```
linetype="dashed",
```

```
size=1)+
```

```
labs(title="Average Monthly Quiz Score",
```

```
x="Month",
```

```
y="Average Quiz Score")+
```

```
theme_minimal()
```

Output

